Daniel Favour Oshidero

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Technical Skills

Languages: Python, SQL, JavaScript (ES6+), Java, VBA
Data & ML: NumPy, Pandas, scikit-learn, PyTorch, spaCy, NLTK, Hugging Face, Neo4j, PowerBI
Cloud & DevOps: AWS, Terraform, Docker, Jenkins, GitHub Actions, Linux (Ubuntu, Amazon Linux)
Web Frameworks: React, React-Native, Angular, Node.js, Flask, HTML/CSS
Additional Competencies: Adobe (Photoshop, Illustrator, Indesign), Autodesk (AutoCAD, Revit, 3DS Max), Unreal Engine
Certifications: AWS Cloud Practitioner, Neo4j Certified Professional, The Complete 2023 Web Developer Bootcamp

Research & Publications

Oshidero, D. F. O. & Coley, D. A. (2026). Talking Carbon: A Lexical Approach to Predictive Carbon Analysis via Machine Learning. *International Journal of Architectural Engineering and Design Management*, pp. 1–34. doi: 10.1080/17452007.2026.2613773

- Introduced a novel ML methodology combining NLP and Histogram-based Gradient Boosting for predicting embodied carbon from conversational design language, validated through real-world case studies and user evaluation.

Relevant Experience

Software Engineer (Full-Stack, AI/ML) — **London, UK**
Intropic *— AI-Driven Finance & Investment Analytics SaaS Provider* — *May 2026 – Present*

- Contribute as member of platform team.

Software Engineer (Data Systems & Infrastructure) — **(Remote) Manchester, UK**
Robert Bosch GmbH *&* ***ETAS Ltd.*** *— Embedded Systems for Automotive Industry* — *Sep. 2024 – May 2026*

- Cut analytics platform costs by ∼88% (**from $5.5k to $675/month**) by replacing managed AWS OpenSearch/OSIS with a self-managed EC2 OpenSearch cluster using Python-based ingesters and parsers, maintaining horizontal scalability with Kubernetes orchestration.
- Designed pattern mining algorithms for sequence-based machine learning models to analyse temporal diagnostic logs and predict next-likely vehicle faults, with approximately **60% accuracy**.
- Architected a production-ready, RESTful NLP interface for Neo4j using transformer intent models and LoRA fine-tuning, enabling non-technical users to query graph database with **above 90% success**.
- Designed and automated system observability for infrastructure operations with Java, Terraform, and Jenkins, managing Linux-based servers (Amazon Linux, Ubuntu) across production and staging environments at scale by monitoring DNS resolution, host connectivity, and service health metrics to generate diagnostic reports for **25+ stakeholders**.
- Implemented AWS security controls and audit-ready logging for compliance under **ISO** and **NIST** frameworks, and **GDPR** requirements (least-privilege IAM, KMS encryption at rest, TLS in transit, and centralised logging via CloudTrail/CloudWatch).
- Re-architected legacy ingestion into a scalable Python XML-to-JSON AWS Lambda pipeline processing **130M+ records/day**, handling high-volume document intake via SFTP/SSH, enabling **over 70% latency reduction** in the pipeline E2E.

Data Engineer & Analyst (BI Infrastructure) — **Wolverhampton, UK**
KTC Edibles Ltd. *— Supplier & Manufacturer for Food & Oil* — *Jun. 2023 – Sep. 2023*

- Engineered an automated database infrastructure by replacing manual CSV workflows with SQL-based ETL pipelines, CRM ingestion scripts, agent scheduling, and validation processes — eliminating **16+ hours** of weekly manual work and enabling **real-time reporting for 80+ employees**.
- Enhanced BI capabilities by implementing row-level security in dashboards, VBA-based data entry capabilities with audit trails, and improved data models — seeing a **93% improvement** in reported user satisfaction and reporting efficiency.

Education

Academic Qualifications

Master of Science, Applied Data Science — **University of Buckingham** — *In Progress.*
Master of Science, Computer Science — **University of Bath** — *Graduated with Distinction*
Awarded the Global Leaders Scholarship for academic excellence and outstanding leadership.
Honours Bachelor of Science — University of Bath — *Graduated with 2:1*

Professional Qualifications

Level 7 Certificate, Digital and Technology Solutions Specialist — *In Progress.*
Level 3 Diploma, Networking and Cybersecurity — *Awarded by Gateway Qualifications*

Projects

Deep RL Models for Increasingly Complex Game Environments: Doom & CartPole | *Python, Pytorch*

* Developed and implemented deep reinforcement learning algorithms using **Markov Decision Process (MDP)** frameworks and **neural networks** (**DDDQN, DRQN, PPO, REINFORCE**) for model training on agents in both simple (CartPole) and complex (VizDoom "Defend the Center") environments.
* Evaluated agent performance across environments using **experience replay**, **target networks**, **reward shaping**, and **exploration vs exploitation** strategies, leveraging **convolutional feature extractors** on raw visual input to learn navigation, resource management, and survival behaviors in high-dimensional state/action spaces.

Bloom: Gamified Plant Learning Application | *JavaScript, React, React-Native*

* Led the design and delivery of **Bloom**, using agile development best practices to build a cross-platform mobile app that gamifies plant care education through interactive learning and habit-forming mechanics.
* Deployed to **12+ physical iOS/Android devices** via Expo for real-world testing, enabling structured **user-interaction data collection** for behavioral analysis and iterative development optimisation.
* Ran A/B tests and weekly feedback loops on onboarding and learning flows, analysing behaviour signals to drive feature development, achieving **80% increase in satisfaction scores** and measurable gains in user engagement.

Simulation of Relational Database Systems (CRUD & Data APIs) | *Python (Tkinter, SQLite3), SQL*

* Built a Python Tkinter airline database application backed by SQLite3, including a designed relational schema and a modular data-access layer exposing CRUD APIs for core entities (flights, employees, airports).
* Implemented queries to database API with input validation, transaction-safe writes, and exception handling to ensure consistent real-time updates.